Modeling 2D Marshak Waves

Optional Subtitle

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Outline

Photon Transport and Energy Exchange

Photon transport (Boltzmann equation):

$$\frac{1}{c}\frac{\partial I}{\partial t} + \hat{\Omega}\cdot\nabla I + (\sigma_a(T_m) + \sigma_s(T_m))I = \sigma_a(T_m)B(T_m) + \frac{\sigma_s(T_m)}{4\pi}\int_{4\pi} I \, d\hat{\Omega} + S(\hat{\Omega}, \vec{r}, t) \quad (1)$$

Energy exchange between radiation and matter:

$$\frac{C_v(T_m)}{c}\frac{\partial T_m}{\partial t} = \sigma_a(T_m)\left(\frac{1}{c}\int_{4\pi} I \, d\hat{\Omega} - aT_m^4\right) \quad (2)$$

Gray Coupled Equations and LTE

We get the "gray" coupled equations:

$$\frac{\partial E}{\partial t} - \nabla \cdot \left(\frac{c}{3\sigma} \nabla E \right) = c\sigma(U - E), \qquad U = \mathrm{a} T_m^4, \quad E = \mathrm{a} T_R^4$$
$$\frac{\partial e}{\partial t} = c\sigma(E - U), \qquad e = f T_m^{\beta} \rho^{-\mu}$$

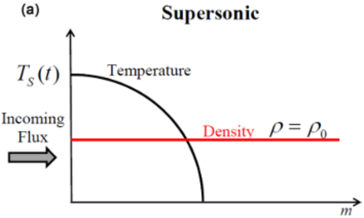
Finally, assuming complete thermodynamic equilibrium (CTE) between the radiation field and the material ($T_R = T_m$):

$$\frac{\partial T^4}{\partial t} + f\rho^{-\mu} \frac{\partial T^{\beta}}{\partial t} = \nabla \cdot \left(\frac{c}{3\sigma} \nabla (aT^4) \right) \tag{3}$$

Assuming $\rho f T^{\beta} \rho^{-\mu} \gg aT^4$ (True until around 3keV):

$$f\rho^{-\mu} \frac{\partial T^{\beta}}{\partial t} = \nabla \cdot \left(\frac{c}{3\sigma} \nabla (aT^4) \right)$$

Motivation



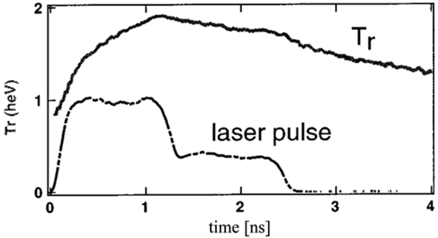
- Given the material properties equations:

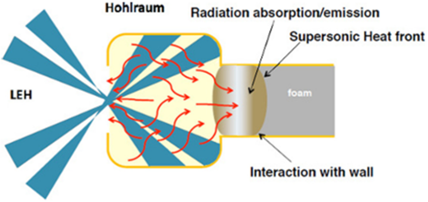
$$e = fT^{\beta}\rho^{-\mu}, \qquad \frac{1}{\kappa} = gT^{\alpha}\rho^{-\lambda}$$

- The diffusion coefficient D is not linear.
- Because of this, the entire system becomes highly non-linear and hard to solve analytically.
- **Our Motivation:** To build a robust model and fundamentally understand the dynamics of this system.

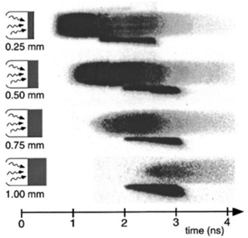
Marshak Waves Experiments

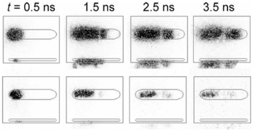
Using a Hohlraum as an x-ray generator, a supersonic heatwave propagates in the low-density foam.

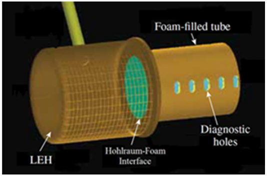




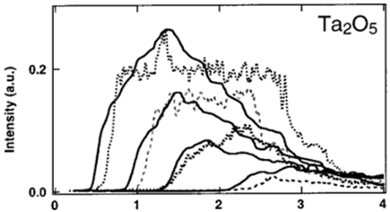
(a)



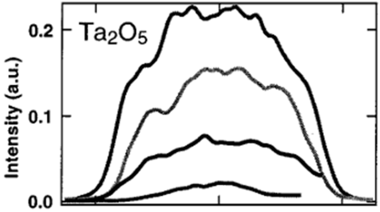




$\Rightarrow$



$\Rightarrow$



Experiments tested

The experiment	foam type	density (mg/cc)	temperature (eV)
Massen *et al.*	$C_{11}H_{16}Pb_{0.3852}$	80	100–150
Back *et al.*	SiO_2, Ta_2O_5	10–50	85–190
Xu *et al.*	C_6H_{12}, $C_6H_{12}Cu_{0.394}$	50	160
Ji-Yan *et al.*	C_8H_8	160	175
Rosen and Keiter *et al.*	$C_{15}H_{20}O_6$, $C_{15}H_{20}O_6Au_{0.172}$	45–70	200–210
Moore *et al.*	SiO_2, C_8H_7Cl	90–120	310
Courtois *et al.*	SiO_2	19 – 29	170

1.5D model

The "Black" diffusion equation (neglecting radiation energy density $\frac{\partial E}{\partial t} \to 0$):

$$\underbrace{\frac{\partial E}{\partial t}}_{\to 0} + \frac{\partial e}{\partial t} = \frac{\partial}{\partial x}\left(\frac{c}{3\rho\kappa}\frac{\partial (aT^4)}{\partial x}\right)$$

With specific energy and opacity models:

$$e = fT^{\beta}\rho^{-\mu}, \qquad \frac{1}{\kappa} = gT^{\alpha}\rho^{-\lambda}$$

where $f, \mu, g, \lambda, \alpha, \beta$ are material-dependent parameters.
HR solved the diffusion equation using perturbation theory and found some interesting results.

HR Model - Heat Front Solution

The heat front position $x_F(t)$ is given by:

$$\boldsymbol{x_F^2(t) = \frac{2+\varepsilon}{1-\varepsilon} C H^{-\varepsilon}(t) \int_0^t H(t')\, dt'}$$

where:

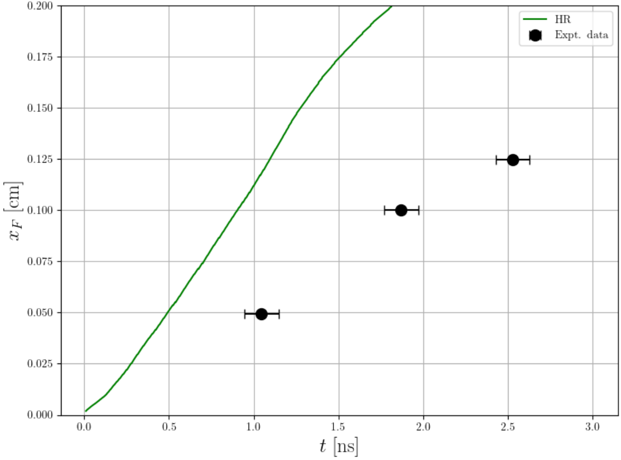
$$C = \frac{16 g \sigma_{SB}}{3f(4+\alpha)} \rho^{2-\mu-\lambda}; \quad \varepsilon = \frac{\beta}{4+\alpha}$$

$$H(t) = T_s^{4+\alpha}(t)$$

where T_s is the surface temperature at $x = 0$.

HR 1D Model Limitations

HR misses the real experimental results of Back et al. by a factor of 2!



1.5D model

1D effects

Marshak Boundary Condition: $T_D \neq T_s$

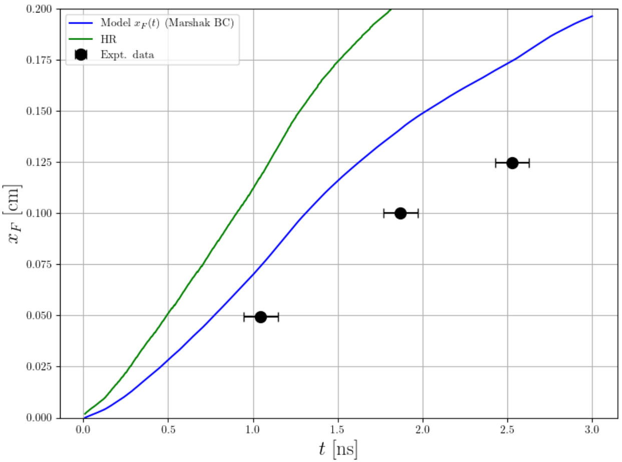
- Assuming $T_s = T_D$ overestimates the wave speed.
- Instead, energy conservation at the boundary requires:

$$\sigma_{SB} T_D^4(t) = \sigma_{SB} T_s^4(t) + \frac{F(0,t)}{2}$$

where $F(0,t)$ is the net flux entering the foam.

Marshak Boundary Condition: $T_D \neq T_s$

Applying the Marshak boundary condition slows the propagation front significantly:



1.5D model

2D effects

Energy Wall Loss

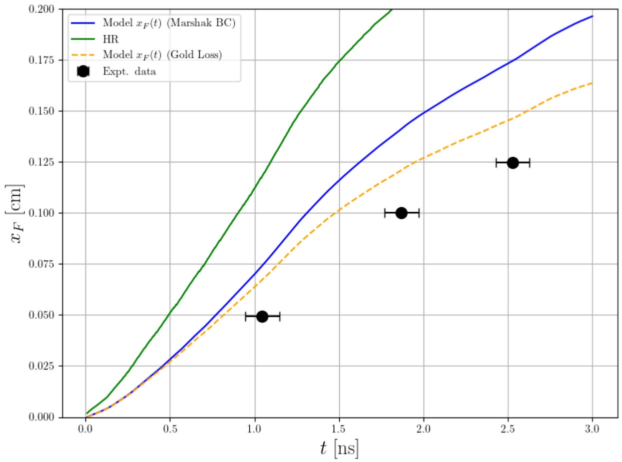
- Energy is lost from the foam into the tube walls.
- Can be modeled for different wall materials: **Gold**, **Copper**, **Beryllium**, and **Vacuum** .
- For **Gold**:

$$E_{\mathrm{gold}}(t) = 0.59\, T_s^{3.35} \left[t - t_0(x)\right]^{0.59} \quad [\mathrm{hJ}/\mathrm{mm}^2]$$

where $t_0(x)$ is the front arrival time at position x.

Energy Wall Loss

Including wall loss reduces the front velocity:

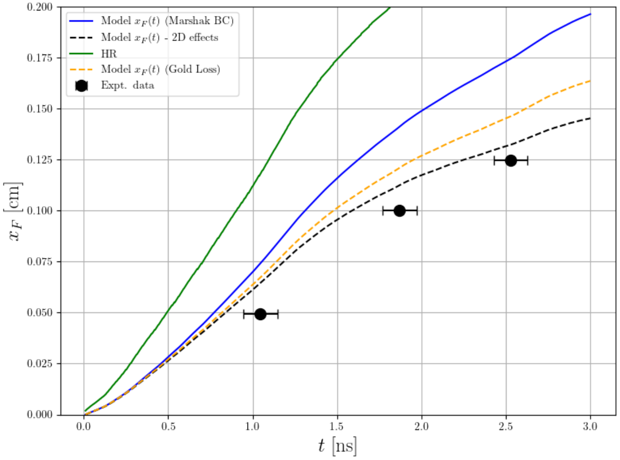


Wall Ablation

- High-Z wall material (e.g. Gold) absorbs energy and ablates inward into the foam.
- **Changing Foam Radius**: The inlet and tube radius shrink, reducing the heat wave cross-section.
- **Changing Foam Density**: The ablated wall material compresses and densifies the adjacent foam.
- Both effects must be incorporated to model the dynamic tube geometry.

Wall Ablation

Inward wall ablation compressively densifies the foam, which significantly slows down propagation:



Effective Mean Free Path

When $\lambda_R \geq 2R$, wall collisions control the transport instead of diffusion. Can be modeled as:

$$\frac{1}{\lambda_{\text{eff}}} = \frac{1}{\lambda_R} + \frac{1}{2R}$$

where λ_R is the Ross Mean Free Path.

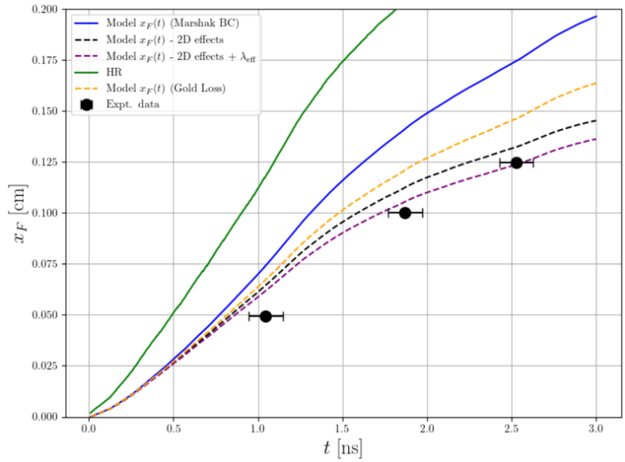
A smoother transition is defined using the generalized power mean:

$$\lambda_{\text{eff}} = \left((2R)^{-n} + \lambda_{\mathrm{R}}^{-n}\right)^{-1/n}$$

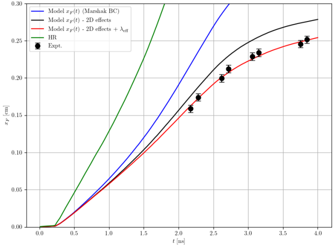
where n controls the "tightness" of the transition.

Effective Mean Free Path

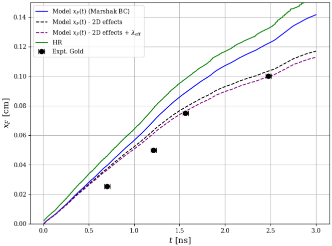
Including the effective mean free path correction matches the simulation to the experimental data at early times:



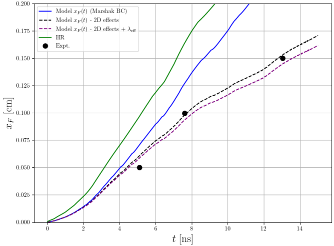
Comparison to many experiments



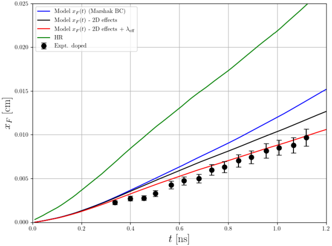
Moore at el. JQSRT, (2015)



Back at el. POP, (2000)



Back at el. PRL, (2000)



Ji-Yan at el. CPB, (2010)

Full 2D model

Modeling the front curvature

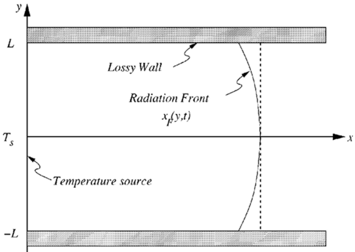
Understanding the front curvature

Hurricane & Hammer (2006) solved the laplace equation (assumed constant opacity and neglected e):

$$\underbrace{\frac{\partial e}{\partial t}}_{\to 0} = \frac{4}{3}\nabla \cdot \left(\frac{1}{\rho\kappa}\nabla(T^4)\right)$$

and got the 2D front in Cartesian coordinates (where $\varepsilon = \frac{3}{4}\rho\kappa L(1-a)$):

$$x_F(y,t) = \frac{L}{\varepsilon}\cosh^{-1}\left(1+\frac{D\varepsilon t}{L^2}\right)\cos\left(\frac{\sqrt{\varepsilon}}{L}y\right) + \text{HOT}$$



Cylinder Coordinates

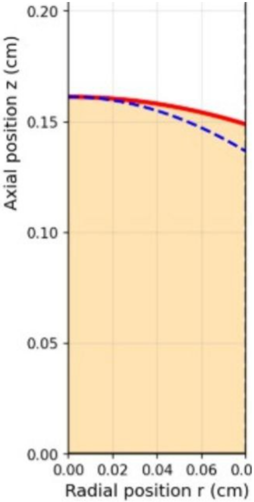
The model was re-developed in cylindrical coordinates (*Courtois at el. POP, (2022)*) to match cylindrical geometries:

$$x_F(r,t) = \frac{1}{\kappa_0}\cosh^{-1}\left(1 + \frac{D\kappa_0^2 t}{2}\right) J_0(\kappa_0 r) + \text{HOT}$$

Planar $\longleftrightarrow$ **Cylindrical Analogy**:
$\cos(ky) \longleftrightarrow J_0(\kappa r)$ For small ε:

$$\kappa_0 \approx \frac{\sqrt{2\varepsilon}}{R}, \quad \varepsilon = \frac{3}{4}\rho\kappa R(1-a)$$

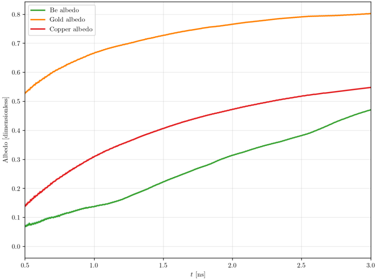
where a is the wall albedo.



Applying on our model: Dynamic Albedo

Until now, wall albedo was treated as a constant parameter. We update it dynamically in time:

$$\alpha(t) = \frac{F_{\text{out, wall}}}{F_{\text{in,wall}}} = \frac{\sigma_{SB} T^4_{s,interface} - \frac{dE_{\text{wall}}}{dt}/2}{\sigma_{SB} T^4_{s,interface} + \frac{dE_{\text{wall}}}{dt}/2}$$



This provides a much more physical description of the energy reflection.

where the curvature goes as $J_0\left(\frac{\sqrt{2\varepsilon(t)}}{R} r\right)$, and $\varepsilon(t)$ scales with $(1-\alpha(t))$.

Problem: the longitudinal front is way off!

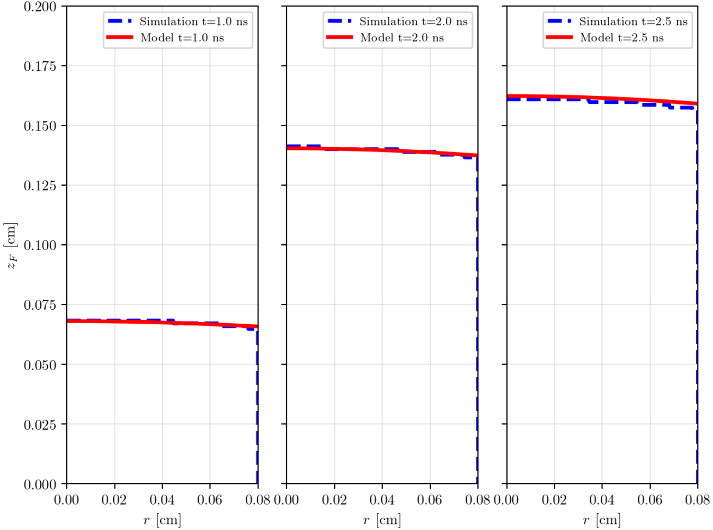
Luckily, we have a direct longitudinal position:

$$x_F(r,t) = x_{1.5\mathrm{D}}(t) \cdot J_0\left(\kappa_0(t)r\right)$$

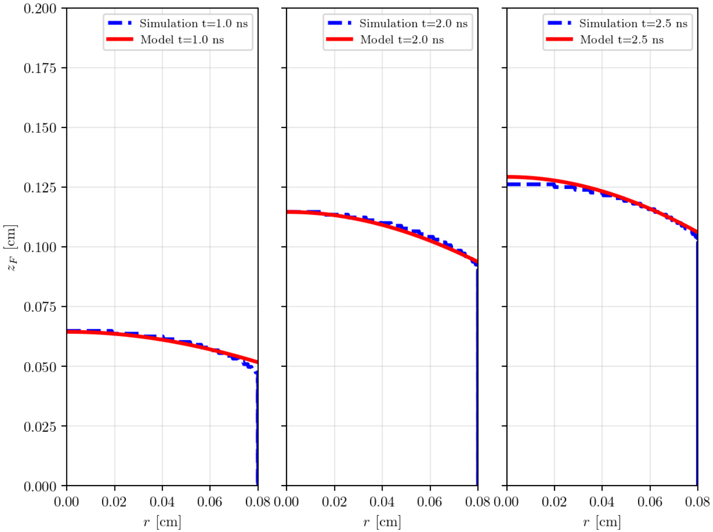
Full 2D model

Comparing to 2D supersonic Diffusion Simulation

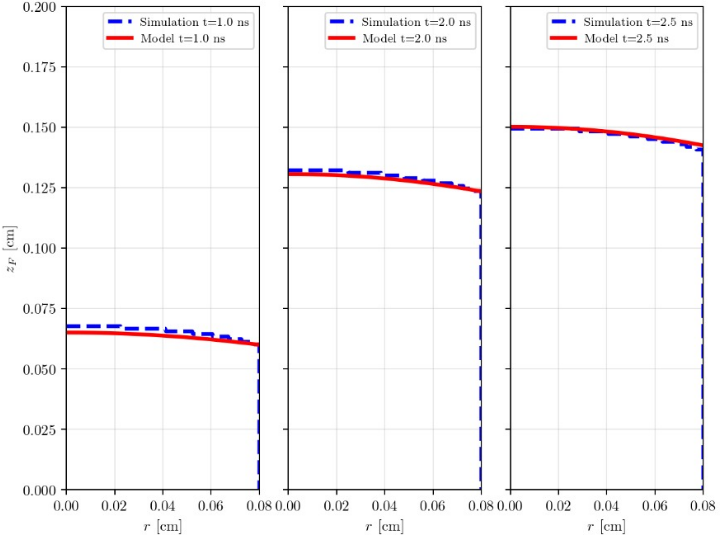
Gold coating - SiO2



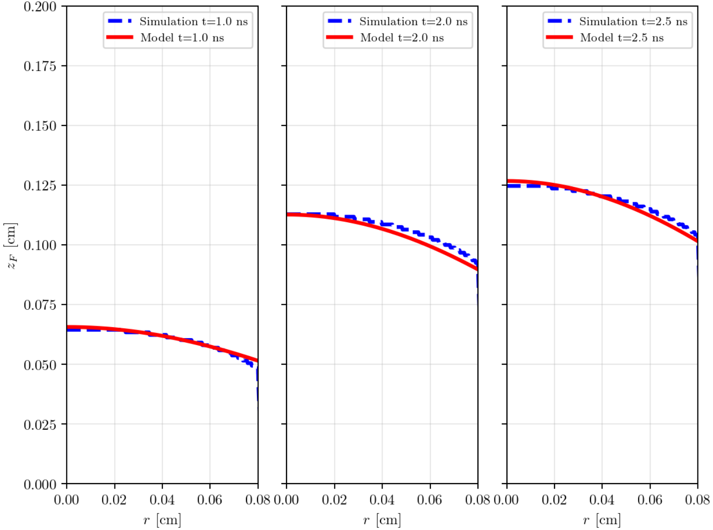
100 times transparenter Gold coating - SiO2



Copper - SiO2

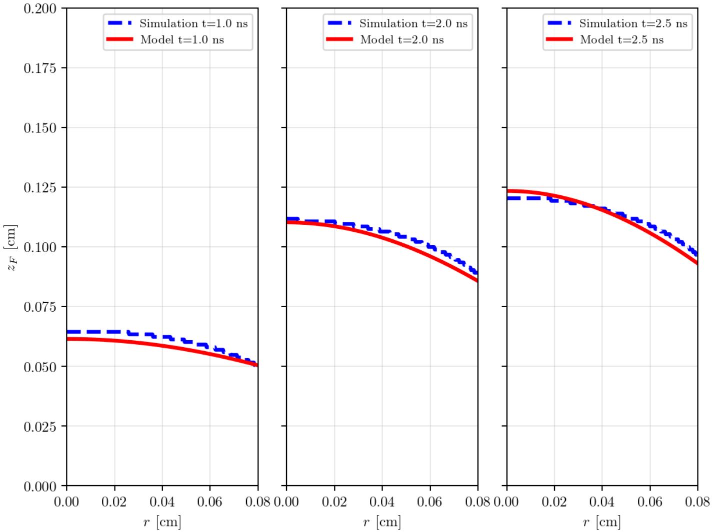

Simulation t=1.0 ns
Model t=1.0 ns
Simulation t=2.0 ns
Model t=2.0 ns
Simulation t=2.5 ns
Model t=2.5 ns
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
z_F [cm]
0.00
0.02
0.04
0.06
0.08
r [cm]

Be - SiO2



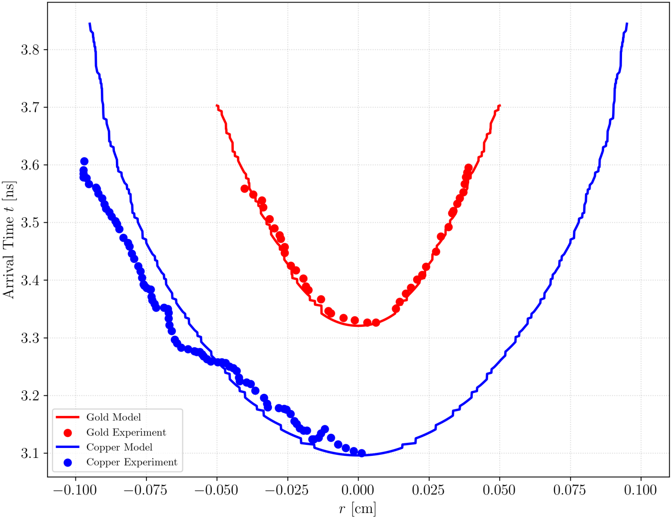
Simulation t=1.0 ns
Model t=1.0 ns
Simulation t=2.0 ns
Model t=2.0 ns
Simulation t=2.5 ns
Model t=2.5 ns
z_F [cm]
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.00
0.02
0.04
0.06
0.08
r [cm]

Vacuum - SiO2


Simulation t=1.0 ns
Model t=1.0 ns
Simulation t=2.0 ns
Model t=2.0 ns
Simulation t=2.5 ns
Model t=2.5 ns
z_F [cm]
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.00
0.02
0.04
0.06
0.08
r [cm]

Gold vs. Copper Coating Comparison

French experiment *Courtois et al. POP, (2024)* which tests the arrival time of the front as a function of r for both gold and copper coats:



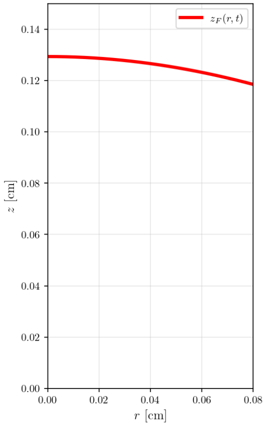
Full 2D model

A full 2D Temperature distribution

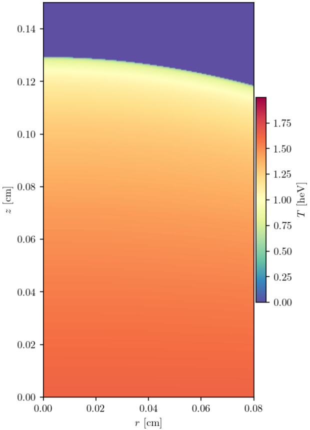
A full 2D Temperature distribution

One can use the Heyney profile:

$$T(z, r, t) \approx T_S(t) \left[1 - \frac{z}{z_F(r, t)}\right]^{\frac{1}{4+\alpha-\beta}}$$



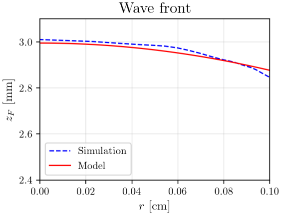
$\Longrightarrow$

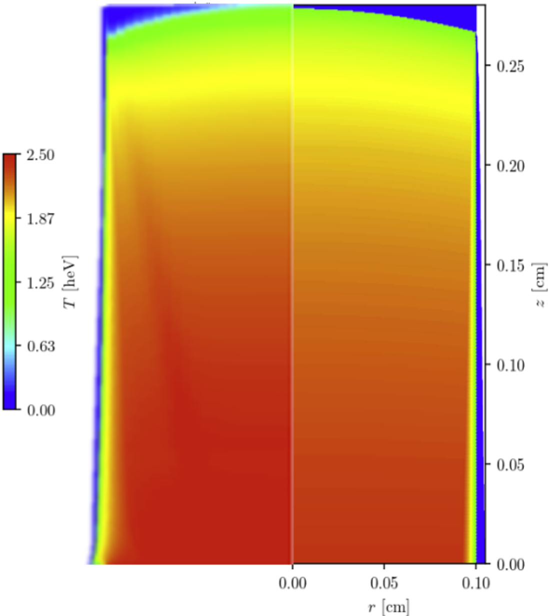


2D Temperature Distribution — Moore (C_8H_7Cl)

Side-by-side comparison of the 2D temperature distribution at $t = 3.72$ ns:

- **Left**: Diffusion simulation taken from *Moore at el. JQSRT, (2015)*.
- **Right**: Our full 2D model.

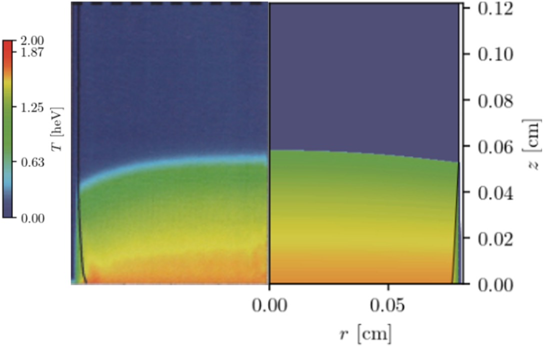




2D Temp. Distribution — Back SiO2 (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

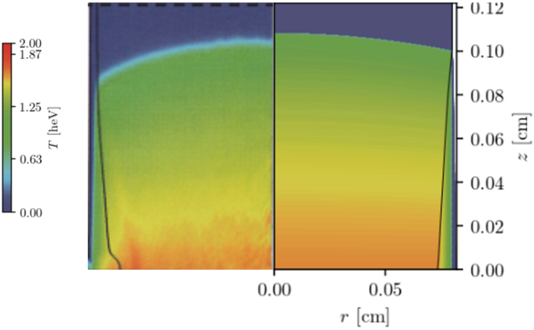
- **Left**: Implicit Monte Carlo (IMC) simulation of *Back et al. POP, (2000)* experiment.
- **Right**: Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temp. Distribution — Back SiO2 (2.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.0$ ns:

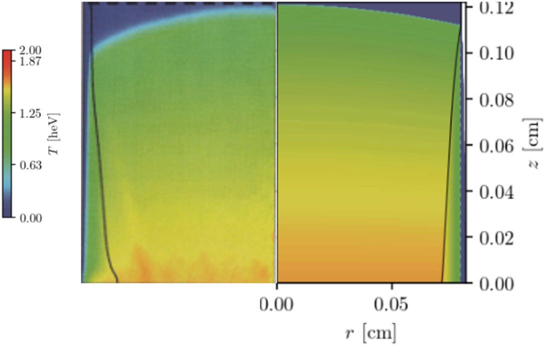
- **Left**: Implicit Monte Carlo (IMC) simulation of *Back et al. POP, (2000)* experiment.
- **Right**: Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.



2D Temp. Distribution — Back SiO2 (2.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 2.5$ ns:

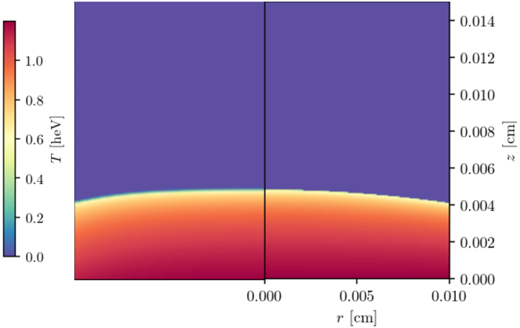
- **Left**: Implicit Monte Carlo (IMC) simulation of *Back et al. POP, (2000)* experiment.
- **Right**: Our full 2D model with $\lambda_{\text{eff}} = 1.5$ correction.

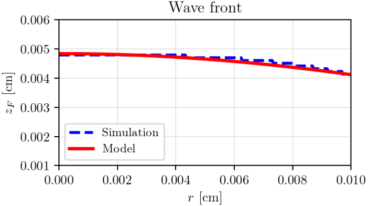


2D Temp. Distribution — Ji-Yan (0.5 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 0.5$ ns:

- **Left**: 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right**: Our full 2D model.

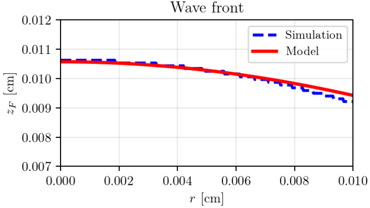


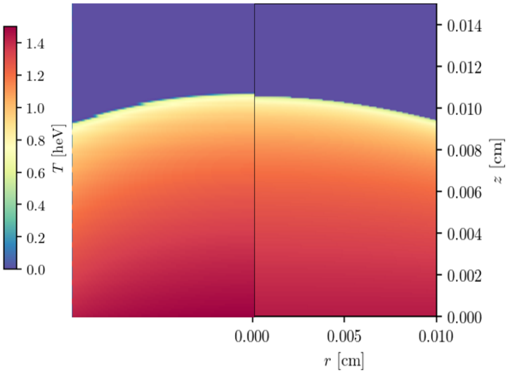


2D Temp. Distribution — Ji-Yan (1.0 ns)

Side-by-side comparison of the 2D temperature distribution at $t = 1.0$ ns:

- **Left**: 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right**: Our full 2D model.

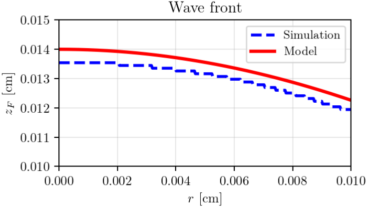


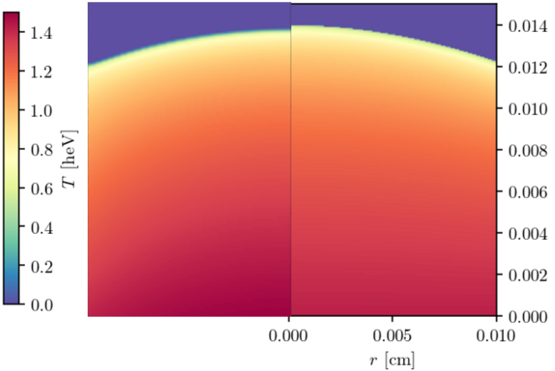


2D Temp. Distribution — Ji-Yan (1.3 ns)

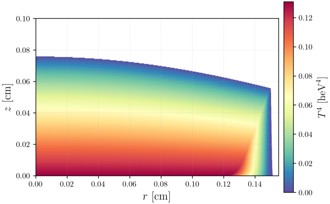
Side-by-side comparison of the 2D temperature distribution at $t = 1.3$ ns:

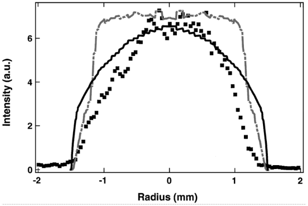
- **Left**: 2D supersonic diffusion simulation of *Ji-Yan et al. CPB, (2010)* experiment.
- **Right**: Our full 2D model.





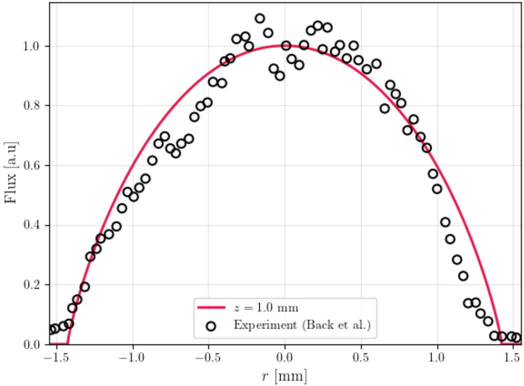
Flux Curvature Comparison - Back SiO2 PRL



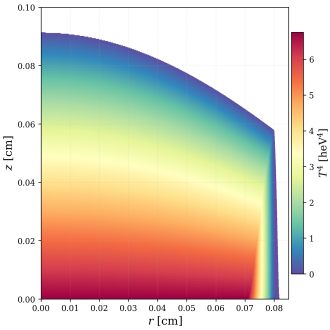


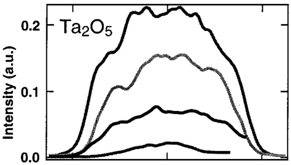
Back at el. POP, (2000)

$\Rightarrow$



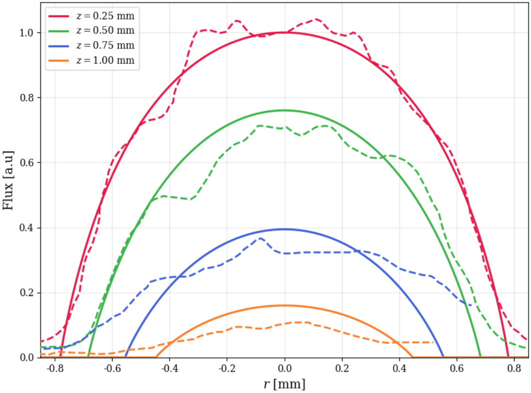
Flux Curvature Comparison - Back Ta2O5 POP





Back at el. POP, (2000)

$\Longrightarrow$



Conclusion

- **Successful 2D Modeling**:
  - We have successfully modeled 2D supersonic Marshak waves.
  - Our unified model captures front curvature and temperature distribution.
- **Deep Physical Insight**:
  - We can now understand the underlying physics of radiation transport and energy exchange in a very good, intuitive way.

Thank You!

Questions?